AMARJEET SINGH

Gurgaon, Haryana | +91 9667743366 | amarjeetsingh4333@gmail.com | Portfolio | LinkedIn | GitHub

SUMMARY

Software Engineer building distributed backend systems in **Golang** and **Python** on **GCP** and **Azure**. Primary author of 3 production systems, including an AI room-mapping platform (10K+ RPS), and led their Azure-to-GCP migration.

SKILLS

Languages/ Frameworks: Golang, Python, SQL, Django, DRF, Flask, REST APIs, OpenAPI/Swagger
Databases/ Messaging: PostgreSQL, Redis, Bigtable, Elasticsearch, MySQL, Kafka, Azure Service Bus, Pub/Sub
Cloud/ DevOps: GCP (Cloud Run, Cloud SQL), Azure (App Service), AWS, Docker, Kubernetes, Airflow, CI/CD
Observability/ AI: Elastic APM, New Relic, Grafana, LLMs (Gemini, OpenAI), Vector Embeddings, Claude Code

WORK EXPERIENCE

Illusions Online Tech India PVT LTD (iOL) – Gurgaon, Haryana
Software Engineer *Oct 2024 – Present*

- Built an AI-powered **room-mapping microservice** (Golang, Redis, PostgreSQL) using **Gemini embeddings** to match rooms across 300+ suppliers (13.8M mappings), serving **10K+ RPS** at **99%+ success**.
- Migrated **3 production services from Azure to GCP** (Cloud Run, Cloud SQL, Pub/Sub, Bigtable, GCS) using cloud-agnostic cache/queue/storage layers and **Azure DevOps CI/CD** pipelines with health-gated deploys.
- Architected event-driven pipelines on **Azure Service Bus** and **Event Hubs** processing **~1M events/day** with concurrent consumers, **dead-letter** reprocessing, and **idempotent** writes; later ported to Pub/Sub.
- Cut dashboard query time **4.5x (6.8s → 1.5s)** and fixed rate-parity queries averaging 34s with a **46% failure rate** via targeted indexes, **table partitioning**, materialized views, and N+1 removal.
- Integrated **7 supplier APIs** (Expedia, Booking.com, Hilton) ingesting **15.6M rooms across 416K hotels**; used **LLMs** to structure raw room data and streamed multi-GB feeds to eliminate OOM crashes.
- Designed a multi-scope **response cache** with ETL-run-aware invalidation and startup prewarming, and migrated it from **Redis to Cloud Bigtable** with per-key TTLs.
- Built **10 of 17 Airflow ETL pipelines** in Golang (bookings, currency, events, demand) into **PostgreSQL**, made crash-safe with resume manifests and **advisory locks**.
- Automated **custom-domain provisioning** for client sites (GoDaddy DNS → **Azure App Service** → SSL).
- Implemented observability with **Elastic APM** distributed tracing and **New Relic** custom metrics and alerts.

Software Intern *June 2024 – September 2024*

- Built room deduplication services using **Python**, cosine similarity, and **Gemini embeddings**, reducing duplicate hotel rooms across suppliers by over 30%; deployed on **Azure Web App**.
- Built **Azure Event Hubs** producers streaming hotel content from 4 suppliers (Expedia, DerbySoft, TGX, Hilton).

Virtuaqaptive Consulting Services PVT LTD – Remote
Python Software Engineer Intern *June 2023 – July 2023*

- Built a **Django** transcription app (**AssemblyAI, OpenAI**) for Meet/Teams/Zoom, deployed on **AWS** with **Docker**.

PROJECTS

RentEase (Python, Django, PostgreSQL, Celery, Redis)

- Built a rental management platform with owner/tenant dashboards, itemized monthly billing, and **Razorpay** payment links with automated email/SMS reminders.
- Processed payment updates via signature-verified, **idempotent webhooks**, with DB constraints preventing duplicate billing; added **Google OAuth** and magic-link login.
- Offloaded **Google Drive** document uploads to **Celery** workers with retries; deployed with **Docker Compose** (Gunicorn, Caddy HTTPS) on DigitalOcean with nightly backups.

UniVerse (Django REST Framework, Azure) – IEEE Published

- Built a college event management platform with user registration, an admin dashboard, and clan points tracking.
- Added OTP password recovery, certificate generation, and soft-delete accounts via Azure-hosted REST APIs.

EDUCATION

Manav Rachna University – Faridabad, Haryana 2021 - 2025
Bachelor of Technology (Computer Science & Engineering): 8.92

CERTIFICATIONS & ACHIEVEMENTS

- IEEE Conference Publication: *"UniVerse: A Comprehensive Solution to College Event Management"* (Nov 2024)
- AWS Academy Graduate – Cloud Architecting, Cloud Computing 101 (Mar 2024)